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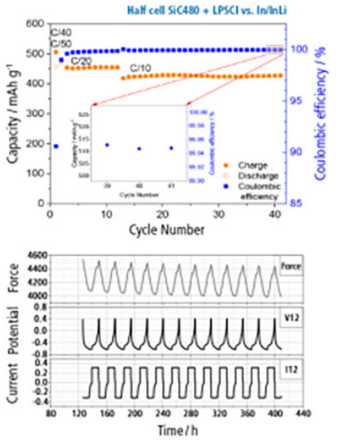
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Impact of Dynamic Driving Loads and Regenerative Braking on the Aging of Lithium-Ion Batteries in Electric Vehicles

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In an electric vehicle (EV), the battery load profiles differ considerably from standard laboratory test procedures, which typically apply constant currents for discharging. Due to the acceleration and deceleration of the vehicle, the battery load is highly dynamic and also contains recharging pulses with higher currents due to regenerative braking. This paper presents an experimental aging study based on a representative driving load profile to investigate battery aging in EVs. The aging study focusses on the impact of regenerative braking on battery aging at different temperatures and SoCs. Moreover, it examines different cycle depths and compares dynamic driving loads to constant current discharging. The study reveals different sensitivities of calendar aging and cycle aging to temperature: Whereas calendar aging decreases with lower temperature, cycle aging increases and also becomes sensitive to the dynamics of the load profile. Cycling up to 200,000 km exhibits that regenerative braking has a beneficial effect on the battery lifetime as it reduces the cycle depth. This lowers the capacity fade and the resistance increase considerably. Furthermore, our paper explains capacity recovery effects occurring after cycling at high SoCs only and presents basic strategies to minimize battery degradation in EVs.

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In electric vehicles (EVs), the load for the traction battery is usually very unsteady during driving. There are dynamic load changes resulting from the acceleration and deceleration of the vehicle. Consequently, the load conditions differ substantially from general laboratory test conditions, where a constant current (CC) discharging is typically performed in cycle life studies (e.g.,[1–4]). Furthermore, driving is not a pure discharging process because regenerative braking leads to recurring recharging periods.

As high charging currents are known to cause accelerated aging due to lithium plating,[5,6] it must be evaluated whether the short recharging pulses owing to regenerative braking lead to accelerated aging. Lithium plating can occur during charging when the anode potential drops below the standard potential of Li/Li^+ due to limitations in charge transfer or lithium solid diffusion.[7,8] Parts of the plated lithium later react irreversibly with the electrolyte, which leads to capacity fade.[9] Graphite anodes are very prone to lithium plating due to their low equilibrium potential, particularly at a high state of charge (SoC).[10] As a general trend, lithium plating increases with higher SoC, higher charging current, and reduced temperature.[5,7]

The impact of dynamic load changes in EVs and of recharging pulses during regenerative braking on battery aging have rarely been examined in the literature. Some studies investigated load profiles for plug-in hybrid electric vehicles (PHEVs).[11–14] However, they did not compare the dynamic pulse profiles to pure CC discharging with the same average current. Furthermore, PHEV operating conditions differ substantially from typical EV operating conditions, particularly in current rates and cycle depth. Several studies on hybrid energy storage systems have recommended the combination of high-energy lithium-ion batteries and supercapacitors to improve cycle life.[15–19] They claim that an additional high-power storage system, which covers all peak loads and recharging pulses, relieves battery stress and, thus, reduces battery degradation. However, no proper validation of this assumption was provided.

Due to the highly complex aging behavior of lithium-ion batteries with its many interdependencies of influencing factors, realistic load patterns should be applied to verify the results from simplified CC cycling procedures.[4] Although there are many publications on cycle aging of lithium-ion batteries, it remains unclear how the frequent load changes during driving operation and, in particular, the recharging pulses during regenerative braking affect the cycle life of lithium-ion batteries in EVs.

To overcome these shortcomings, our paper presents an experimental aging study focusing on the impact of regenerative braking on battery aging at different SoCs, temperatures, and cycle depths. Moreover, the aging owing to the dynamic driving load profile is compared to pure CC discharging. Furthermore, capacity recovery effects after longer inactivity periods are identified and analyzed.

Experimental

To examine battery aging for an electric passenger vehicle, an experimental aging study was conducted with a representative driving load profile. As battery aging is cell-specific and strongly dependent on the cell design and the cell chemistry, an appropriate cell type had to be selected to obtain results that can also be transferred to other cells used in EVs. This section presents the cell type examined, the load profile, the operating conditions, and the overall test procedure.

Cell type.—In our aging study, cylindrical cells of type Panasonic NCR18650PD were examined. These high-energy cells feature a nominal capacity of 2.8 Ah, a specific energy of 220 Wh/kg, and an energy density of 570 Wh/l. They contain a graphite anode and a lithium nickel cobalt aluminum oxide (NCA) cathode. A low internal resistance of approx. 20 mΩ, a good capacity utilization at elevated discharge currents, up to 10 A, and a good discharge performance at low temperature, even below 0°C, qualify these cells for examining EV operating conditions.

Load profile.—Instead of discharging with a constant current or constant power, a dynamic driving load profile was employed. To examine the impact of regenerative braking on battery aging, a suitable driving load profile had to be selected for the study. As described in Ref. 20, we computed various load profiles from velocity profiles of driving cycles used for fuel economy measurements. These computations were performed with a vehicle model containing the different driving resistances. A passenger vehicle configuration providing only a 100 km driving range was used to obtain demanding battery load profiles. From different load profiles,[21] the American US06 driving cycle was selected for this aging study, as it represents aggressive highway driving with frequent changes between acceleration and deceleration. This led to more recharging intervals and higher recharging currents than for other driving profiles. Applying constantly the US06 highway driving profile represents a worst-case usage scenario, as EVs are often used in urban traffic conditions, where, despite of more frequent stop-and-go driving conditions, the battery load is

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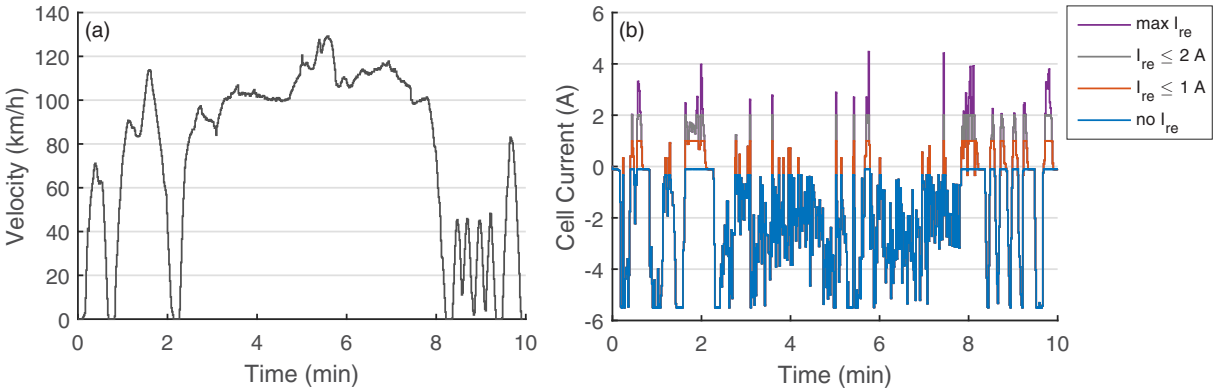


Figure 1. (a) Velocity profile of the US06 highway driving cycle. (b) Load profiles for a single lithium-ion cell with different magnitudes of regenerative braking, represented by different limitations for the recharging currents (I_{re}) during regenerative braking.

considerably lower.[20] The velocity profile of the US06 driving cycle is shown in Figure 1a. Figure 1b exhibits the corresponding charging and discharging currents for a single lithium-ion cell, computed in Ref. 20. As the cycle aging study was conducted on a battery test system with 5 V/±5 A test channels, high discharging loads had to be truncated at −5.5 A, which was the current limit of the test system.

Variation of regenerative braking.—To analyze the impact of regenerative braking on battery aging, four different current levels were defined for the recharging pulses, which are illustrated in Figure 1b by the four different load profiles: The first level is "no recharging" ("no I_{re}"), which corresponds to no regenerative braking at all. The load during the periods of regenerative braking or standstill remains somewhat below 0 A, since auxiliary loads still have to be provided. The second level limits recharging currents I_{re} for the battery cell to 1 A ("$I_{re} \leq 1$ A"). It can recover 8% of the ampere-hours discharged during the driving cycle, which corresponds to approximately half of the maximum recoverable amount of charge. The third level of regenerative braking was defined as "$I_{re} \leq 2$ A", covering more than 80% of the maximum recoverable amount of charge. The fourth level of unrestricted regenerative braking is labeled "max I_{re}" and contains recharging pulses up to 4.5 A. This enabled a charge recovery of 16%.

State of charge levels.—One run of the US06 driving cycle corresponds to a driven distance of about 13 km. To obtain more representative driving distances for the cycle life study, at least two runs of the load profile were performed in series before charging the cell again. Without regenerative braking, two consecutive runs of the US06 highway driving cycle depleted approximately one fourth of the cell's nominal capacity C_N.

In the cycle aging study, different SoC levels were examined. The SoC levels were defined by the maximum voltage for the CC charging procedure with a charging current of 0.7 A (= C/4). As illustrated in Figure 2, the low, medium, and high charging voltages were 3.7 V, 3.9 V, and 4.1 V, respectively. A slight correction of charging voltages was applied for 10°C: As higher internal resistances at low temperatures lead to higher terminal voltages during charging, the low charging voltage level was set to 3.75 V and the medium charging voltage was set to 3.925 V. The high charging voltage remained at 4.1 V to avoid any risk of increased lithium plating.

Cycle depths.—The double arrows in Figure 2 illustrate the cycling windows for two subsequent runs of the driving load profiles. Two consecutive runs of the driving load profile deplete 24.8% C_N. This corresponds to a depth of discharge of 24.8% C_N when no regenerative braking is used. By recovering charge during braking periods, the depth of discharge at the end of the driving sequence remains smaller. With unrestricted regenerative braking, two consecutive runs of the driving load profile lead to a depth of discharge of 20.4% C_N.

In addition to two consecutive runs of the driving load profile, four and six consecutive runs were also examined for the driving load profile with maximum regenerative braking. This depleted 49.5% and 74.3% C_N. Due to the partial recharging by unrestricted regenerative braking, the resulting depths of discharge were 40.8% and 61.1% C_N, respectively. These higher cycle depths were examined for the high and the medium charging voltage.

The remaining SoC margins toward 0% and 100% SoC guaranteed that no overcharging occurred during recharging pulses at high SoC and allowed to perform the cycling procedure also on aged cells with lower capacities approaching the end of life condition of 20% capacity fade.

Cycle aging vs. calendar aging.—The aging of lithium-ion batteries generally comprises usage-dependent cycle aging and usage-independent calendar aging.[22] Both lead to a loss of capacity and an increase of internal resistances. Although battery aging is highly complex, it is often assumed that the contributions of both aging processes add linearly.[23,24]

To separate cycle aging and calendar aging in this study, baseline curves for calendar aging were required. Therefore, additional cells were stored at different temperatures and SoCs to quantify calendar aging. Figure 2 exhibits the different storage SoCs to which the cells were charged with a CC charging procedure with 700 mA. Figure 2 also illustrates from which two storage SoCs, an average calendar aging baseline curve was computed for the three cycling windows

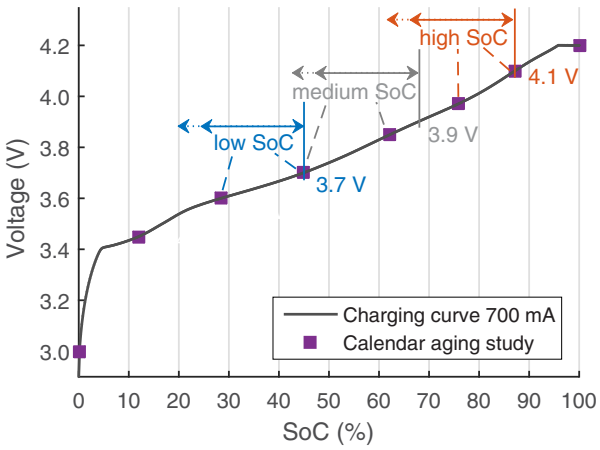


Figure 2. Cycling windows for low, medium, and high charging voltage for constant-current charging at 25°C. The double arrows illustrate the cycle depths for two consecutive runs of the driving load profiles with no regenerative braking at all (25% C_N) and maximum regenerative braking (20% C_N). Moreover, the SoCs examined for calendar aging are illustrated.

Table I. Overview over the test conditions examined to investigate the impact of different influencing factors on battery aging in an EV.

Research Objective	Load Profile	Regenerative Braking	Cycle Depth (C_N)	Charging Voltage (V)	Temperature (°C)
Impact of temperature	US06 cycle (2x)	no I_{re}	24.8%	4.1, 3.9, 3.7	40, 25, 10
Impact of regenerative braking	US06 cycle (2x)	no I_{re}	24.8%	4.1, 3.9, 3.7	40, 25, 10
	US06 cycle (2x)	$I_{re} \leq 1$ A	22.3%	4.1, 3.9, 3.7	40, 25, 10
	US06 cycle (2x)	$I_{re} \leq 2$ A	21.1%	4.1, 3.9, 3.7	40, 25, 10
	US06 cycle (2x)	max I_{re}	20.4%	4.1, 3.9, 3.7	40, 25, 10
Impact of cycle depth	US06 cycle (2x)	max I_{re}	20.4%	4.1, 3.9, 3.7	40, 25, 10
	US06 cycle (4x)	max I_{re}	40.8%	4.1, 3.9	40, 25, 10
	US06 cycle (6x)	max I_{re}	61.1%	4.1	40, 25, 10
Long-term degradation (2000 EFC)	same as for the impact of cycle depth, but 25°C only				
Dynamic vs. constant discharging	US06 cycle (4x)	max I_{re}	40.8%	4.1	40, 25, 10
	CC discharging	–	40.8%	4.1	40, 25, 10

resulting for two subsequent runs of the driving load profile. Detailed insights into the calendar aging behavior of the examined lithium-ion cells can be found in Refs. 25,26.

Temperature conditions.—To examine the impact of different operating temperatures on battery aging, cell storage as well as the cycling sequences with the driving load profiles were performed in thermal chambers at different temperatures of 40°C, 25°C, and 10°C.

For the periodic checkup measurements, monitoring the degradation progress, all cells were placed in thermal chambers at 25°C to obtain comparable results.

Checkup.—The entire aging study was performed with a 32-channel BaSyTec CTS battery test system. Periodic checkups were performed to track the capacity fade and the resistance increase of the cells.

To determine the capacity fade, the cells were fully charged with a CCCV charging protocol (I_{ch} = 700 mA, U_{ch} = 4.2 V, I_{end} = 100 mA). After that, the cells were CCCV discharged (I_{dis} = –3 A, U_{dis} = 2.75 V, I_{end} = –100 mA) to obtain the actual capacity.

At the end of the checkup, the cells were charged to 50% SoC. At this SoC, current pulses and electrochemical impedance spectroscopy (EIS) were applied to detect changes in the internal resistances. From a –3 A discharge pulse, the actual $R_{dc,10s}$ was computed based on the cell voltage right before the current pulse and after 10 s of current flow. This value contains resistance contributions from ohmic conductors, electrolyte conductivity, passivation layers, charge transfer, and diffusion.[27,28]

Moreover, a galvanostatic EIS measurement ranging from 10 kHz to 10 mHz was performed with a GAMRY G750 galvanostat/potentiostat and an RMS excitation amplitude of 50 mA. The real part of the impedance at 1 kHz is presented as $R_{ac,1kHz}$. This high-frequency resistance value is used to evaluate changes in the ohmic cell resistance, owing mainly to changes in the electrolyte conductivity.[29–31]

Test procedure.—At the beginning of the cycle aging study and every 100 equivalent full cycles (EFC), a checkup was performed at 25°C to track the degradation of the cells. 1 EFC represents the charge throughput of 1 C_N charged and 1 C_N discharged. Due to the vehicle configuration employed, 1 EFC corresponds to a driven distance of about 100 km.

Between two checkup measurements, each cell was put into the respective thermal chamber and underwent the cycling period. The cycling procedure combined CC charging with 700 mA to the respective charging voltage and discharging sequences with the respective load scenario. The low charging current of C/4 was employed to minimize degradation caused by charging the cells. For discharging the cells, the driving load profiles with four different magnitudes of regenerative braking were used. Two, four, or six consecutive runs of the load profiles, with a pause of 1 min in between, were applied to obtain different cycle depths. A pause of 5 min was inserted between each charging and discharging sequence. In total, a cycling sequence of 100 EFC had a duration of about 4 weeks.

Table I summarizes all test conditions and shows which load cases were evaluated for which research objective.

Results and Discussion

To investigate the impact of driving operation with dynamic loads and regenerative braking on battery aging, up to 2000 EFC were performed for different operating temperatures, SoCs, levels of regenerative braking, and cycle depths. For the vehicle configuration examined (see Ref. 20), this corresponds to a driven distance of up to 200,000 km.

Impact of temperature.—Figure 3 illustrates the degradation of the lithium-ion cells when the cells were charged with 700 mA and discharged without regenerative braking at 40°C, 25°C, and 10°C. As there was no charge recovery during the driving sequences, the cycle depth was approximately 25% C_N. Figure 3 shows the impact of SoC and temperature on capacity fade and resistance increase. To distinguish between usage-independent and usage-dependent battery degradation, baseline curves for calendar aging are included in the capacity plots.

Capacity fade.—The capacity fade for the three operating temperatures exhibits a strong dependency on SoC. After 500 EFC, corresponding to a driven distance of about 50,000 km, the capacity fade amounts to approx. 5.5% at low SoC, to 6.3–7.2% at medium SoC, and to 8–12% at high SoC. Thus, a higher SoC has led to faster degradation. As this trend can originate to a certain extent from usage-independent calendar aging, the individual contributions of calendar and cycle aging must also be regarded separately.

Figures 3a–3c demonstrate that the contributions of calendar and cycle aging vary substantially with temperature. For the cells tested at 40°C, calendar aging has a considerable impact on the overall degradation and is responsible for more than half of the capacity fade. The additional capacity fade owing to cycle aging amounts to 2.5% for low and medium SoC, and to 4% for high SoC. For the cells cycled at 10°C, the highest capacity fade is observed at all SoC levels. Since calendar aging is low at this operating temperature, the capacity fade is caused predominantly by cycle aging. The contribution from cycle aging, in addition to the 1–2.5% of capacity fade from calendar aging, amounts to 4.5% at low SoC, to 5.5% at medium SoC, and to 9.5% at high SoC. Overall, the combination of low temperature and high SoC exhibits the largest capacity fade. The lowest aging has been obtained at 25°C. With a capacity fade of 1.5–3%, calendar aging at 25°C is

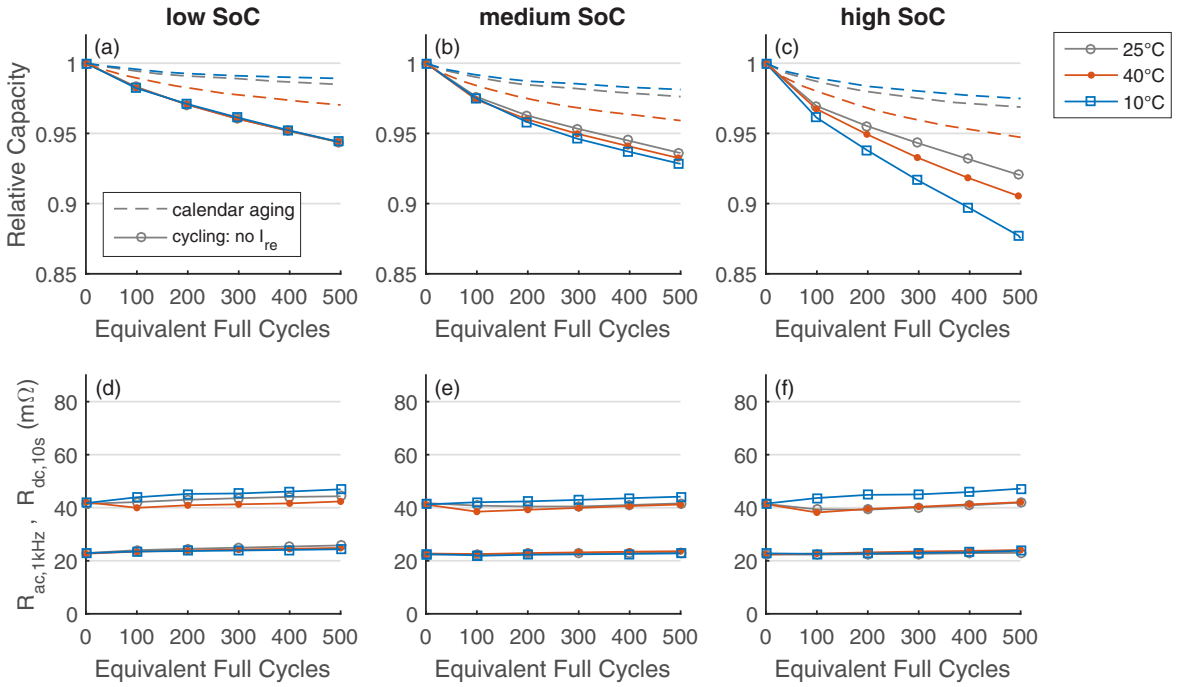


Figure 3. (a-c) Capacity fade and (d-f) resistance increase for 500 EFC (about 50,000 km) without regenerative braking and with a cycle depth of 25% C_N. For the usage-independent capacity fade, the dashed calendar aging curves are scaled in accordance with the 4 weeks of cycling between two consecutive checkups.

considerably lower than at 40°C. The additional capacity fade owing to cycle aging amounts to 4% at low and medium SoC, and to almost 5% at high SoC. Thus, cycle aging – as the difference between the overall degradation and pure calendar aging – is more severe than at 40°C, but lower than at 10°C.

Due to the different impact of temperature on calendar and cycle aging, there is no steady increase or decrease of the entire capacity fade with temperature. The capacity fade increases toward higher and lower temperatures. 25°C has provided the best cycle life in this aging study. With higher temperatures, calendar aging increases whereas at low temperatures, the cycling operation causes substantial degradation. These results are in good agreement with other aging studies, which reported an optimal operating temperature around 25°C and increasing degradation toward higher and lower operating temperatures.[32,33] For higher temperatures, Arrhenius-driven reactions were reported as main driving forces of aging, which strongly correlates with calendar aging. For lower temperatures, degradation was ascribed to aggravating lithium plating.

Resistance increase.—In addition to the capacity fade, battery aging leads to rising internal resistances. Figures 3d–3f show the resistance changes for the different operating temperatures and SoC levels. The increase in $R_{ac,1kHz}$, representing the ohmic behavior, lies below 3 mΩ after 500 EFC for all cells, which means below 15%. The changes in $R_{dc,10s}$ of up to 5.6 mΩ also lie below 15% after 500 EFC, where the largest resistance increase occurs at 10°C. Overall, the resistance changes remain rather small and do not exhibit strong dependencies on the operating SoC, as it has been observed for the capacity fade.

In the beginning of the cycling study, $R_{dc,10s}$ has even decreased for all cells cycled at 40°C and for the cells at 25°C that were cycled at the high or medium SoC level. EIS measurements revealed that this resulted from shrinking resistance contributions from the high-frequency capacitive semicircle, typically ascribed to the SEI at the anode. Lower anode resistances after cycling were also reported in several studies in the literature.[34–36]

Impact of regenerative braking.—For the test procedure with two consecutive runs of the US06 highway driving profile, the impact of regenerative braking on battery aging was examined with four different levels of maximum recharging currents during the braking periods. The four levels ranged from no recharging at all to unlimited recharging pulses. For 500 EFC, representing ca. 50,000 km, the impact of regenerative braking on battery aging is analyzed.

Capacity fade.—For the nine combinations of operating temperature and SoC level, Figure 4 depicts the capacity fade for the four levels of regenerative braking. Again, the capacity fade is compared to calendar aging to identify usage-dependent and usage-independent aging contributions.

At 40°C, the cells cycled at the low and medium SoC level show almost no dependency on the magnitude of regenerative braking. All curves lie closely together in Figure 4a and Figure 4b. After 500 EFC, the cells cycled at low SoC exhibit a capacity loss of almost 6% and the cells cycled at medium SoC of almost 7%. The additional capacity loss at medium SoC can be ascribed to increased calendar aging. Thus, cycle aging in addition to the capacity loss from calendar aging is considered identical for both SoC levels and amounts to ca. 2.5% capacity fade after 500 EFC. Regarding the high SoC level at 40°C, Figure 4c demonstrates a dependency on regenerative braking as the capacity curves diverge. A trend becomes apparent: A higher level of regenerative braking correlates with a reduced capacity loss. After 500 EFC, the capacity loss in addition to calendar aging is ca. 2.5% for the cell with unrestricted regenerative braking and more than 4% for the cell with no regenerative braking.

In contrast to 40°C, Figures 4d–4f show a dependency of battery aging on the level of regenerative braking for all three SoC levels at 25°C: Cells with a higher magnitude of regenerative braking exhibit lower capacity losses. In analogy to 40°C, the capacity fade increases with higher SoC level. At low and medium SoC, the capacity fade from cycle aging is comparable for the same levels of regenerative braking. At high SoC, intensified aging can be observed again: Capacity losses due to cycling are more than 1 percentage point higher than at medium and low SoC. Compared to 40°C, calendar aging is about 2 percentage points lower and cycle aging is about 1 percentage point higher at 25°C.

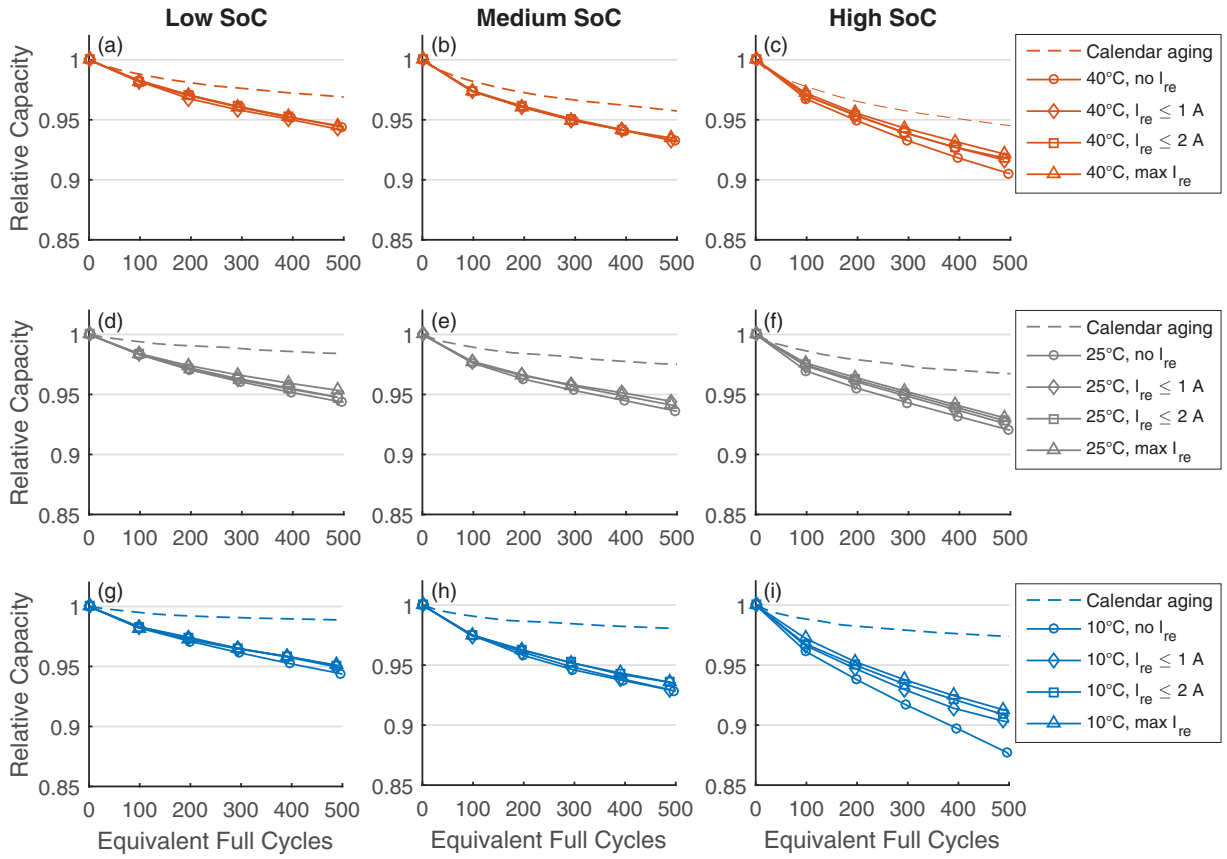


Figure 4. Capacity fade for 500 EFC (about 50,000 km) with different magnitudes of regenerative braking. Two subsequent runs of the driving load profile were performed before each recharging. The calendar aging curves are scaled in accordance to the about four weeks of cycling between two consecutive checkups.

At 10°C, Figures 4g–4i again present a dependency on the level of regenerative braking for all SoC levels. The most pronounced impact can be observed at the high SoC level, where capacity curves diverge markedly. Although calendar aging is low at 10°C, the cells exhibit substantial capacity losses. After 500 EFC, capacity losses owing to calendar aging amount to 1% for low SoC, 2% for medium SoC, and 2.5% for high SoC. For the low and medium SoC level, additional losses of 4–5% can be ascribed to cycle aging. At the high SoC level, the additional capacity fade owing to cycle aging amounts to 6% for the cell with maximum regenerative braking. For the cell with no regenerative braking, elevated cycle aging of 10% capacity fade results in a total capacity fade of 12.5% after 500 EFC. This represents the most severe degradation of all load cases depicted in Figure 4.

Analyzing the capacity fade with respect to regenerative braking reveals a common trend: Reduced battery aging goes along with a higher level of regenerative braking. This trend has been observed most pronouncedly for the high SoC and low temperature. As these operating conditions are those for which strong degradation owing to lithium plating was expected, the concern of regenerative braking causing more lithium plating in EVs is not affirmed. This result is in good agreement with another recent study on lithium plating, which confirmed that lithium-ion cells can tolerate substantially higher charging currents during short charging pulses than during prolonged charging sequences.[37]

Resistance increase.—The resistance increase of $R_{ac,1kHz}$ and $R_{dc,10s}$ did not reveal a dependency on regenerative braking. The results for all four levels of regenerative braking were similar to those presented in Figures 3d–3f for the cells cycled without regenerative braking. A decrease in $R_{dc,10s}$ at the beginning of the cycling procedure occurred for all cells cycled at 40°C and for the cells cycled at 25°C with medium and high charging voltage. As the resistance changes were rather small, the capacity fade turned out to be the predominant aging effect for all the cells cycled with the low cycle depth resulting from two consecutive runs of the driving load profile.

Effects of reduced cycle depth.—The lower capacity fade from cycle aging with higher levels of regenerative braking can be explained by the reduced depth of discharge after the driving sequences: For the four load profiles with different magnitudes of regenerative braking, the overall charge throughput is similar. They all discharge the same amount of ampere-hours during the two subsequent runs of the US06 highway driving load profiles. Only the time points of recharging differ due to the microcycles generated by regenerative braking. With higher magnitudes of regenerative braking, more ampere-hours are recharged during the driving sequences and the cells are less discharged after the driving sequences. Thus, the overall cycle depth decreases and the subsequent CC charging period shortens.

To analyze the impact of different cycle depths, Figure 5 depicts the capacity fade with respect to "equivalent full cycles without microcycles". This excludes the charge throughput of the microcycles resulting from charge recovery during braking periods. As a consequence, this regards only the charge throughput of the main discharge-recharge cycle, which corresponds to the ampere-hours recharged during the CC charging period after the driving sequence.

As expected, Figure 5 shows that the EFC without microcycles are considerably lower for cells with higher levels of regenerative braking. For cells with maximum regenerative braking, the cycle number reduces from 500 to 420, whereas it remains unchanged for cells without regenerative braking. Comparing Figure 4 and Figure 5, the adjustments owing to the different scale bases become apparent: At 10°C and 25°C, capacity curves lie closer together in Figure 5 for all SoC levels. At 40°C, this effect can also be observed at high SoC. This

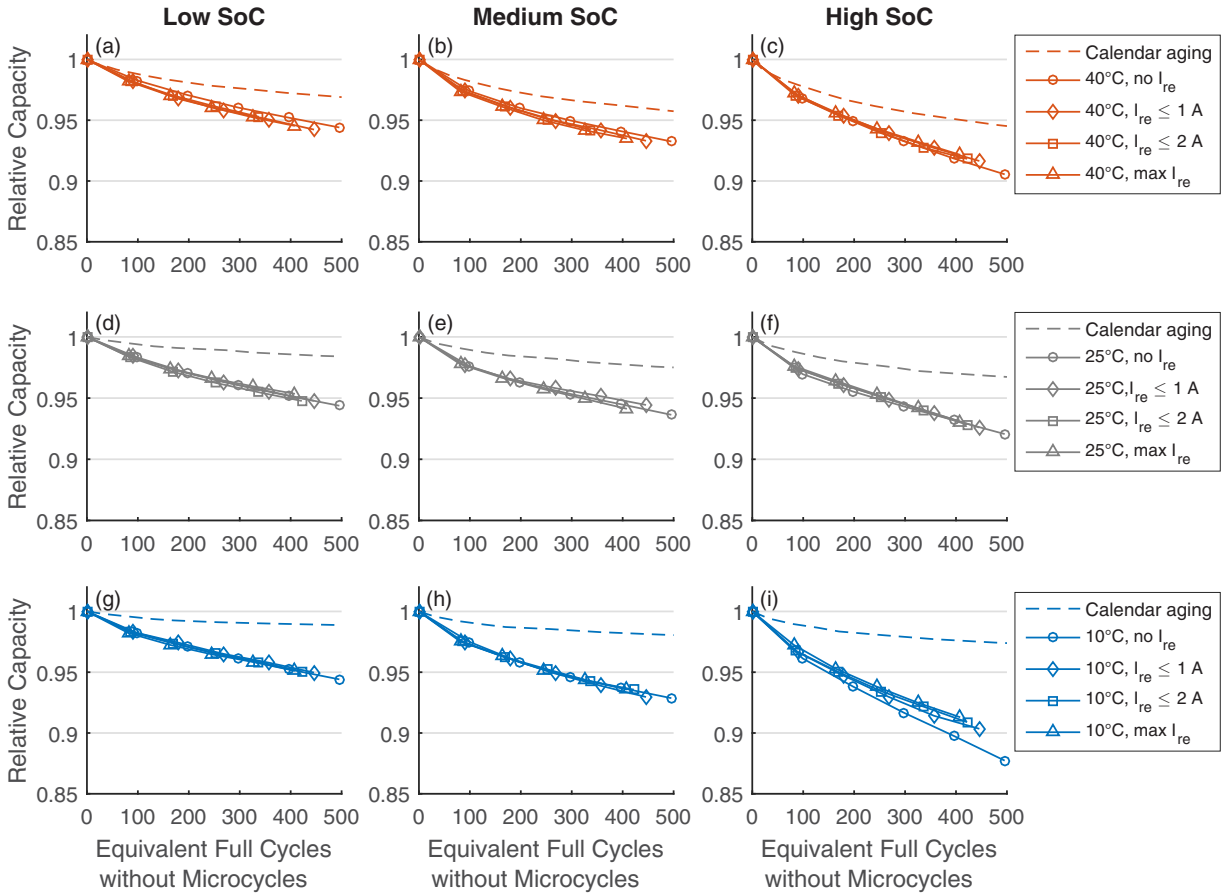


Figure 5. Capacity fade versus EFC without microcycles, which excludes the charge throughput from partial recharging by regenerative braking. Same data as in Figure 4, but with a different values for the x-axis. Scaling of calendar aging identical as in Figure 4.

confirms the direct correlation between changes in cell degradation and changes in cycle depth under most operating conditions.

As high SoC and low temperature aggravate lithium plating and as decreasing cycle depths with higher magnitudes of regenerative braking have reduced cycle aging particularly under these operating conditions, it is supposed that the lower cycle depth has reduced lithium plating. As the capacity curves lie closer together when the charge throughput of the microcycles resulting from regenerative braking are neglected, it is concluded that the CC charging sequence could be the driving force of capacity fade as it caused small amounts of lithium plating. Consequently, it is not the short recharging with high current rates during regenerative braking that promotes lithium plating, but the long-lasting charging periods when the vehicle is recharged by the grid, although current rates are considerably lower.

Only for the low and medium SoC level at 40°C, the correlation between cell degradation and EFC is in better agreement. Under these operating conditions, lithium plating is unlikely to occur for the charging currents applied in this aging study. As Figure 4a and Figure 4b show, even under these operating conditions, regenerative braking has not intensified battery aging. Thus, a high level of regenerative braking appears beneficial to the cycle life of a lithium-ion battery in an EV.

Impact of cycle depth.—The results presented in the preceding sections have comprised 500 EFC for cycle depths between 20% and 25% C_N. To further examine the impact of cycle depth, larger cycle depths were tested by test procedures including four and six – instead of two – consecutive runs of the driving load profile. As regenerative braking had been identified as beneficial for the cycle life of the lithium-ion cells, the load profile with unrestricted regenerative braking was used in the test procedures for comparing different cycle depths. Figure 6 illustrates the capacity fade and the resistance increase for the three operating temperatures and six load scenarios with different charging voltages and cycling depths. It presents up to 1200 EFC, which corresponds to a driven distance of about 120,000 km.

Capacity fade.—Figure 6a illustrates that the capacity fade after 1200 EFC at 40°C ranges between 9% and 18%. The lowest capacity fade is observed for the low cycle depth of 20% C_N in combination with the low charging voltage. The highest capacity fade is observed for the largest cycle depth of 61% C_N. For the medium cycle depths of 41% C_N, a higher capacity fade has occurred than for the cells with a cycle depth of 20% C_N which have been charged to the same charging voltage.

For an operating temperature of 25°C, Figure 6b reveals capacity fades between 7.5% and 12%. Again, the capacity fade increases with larger cycle depths and higher charging voltages. In the beginning of the aging study, the two cells charged to 4.1 V and discharged with cycle depths of 41% and 61% C_N exhibit a similar capacity fade. After ca. 300 EFC, however, the slope of the capacity curve of the cell with 41% C_N cycle depth flattens and the rate of the capacity fade becomes even lower than for the cell cycled at high SoC with a cycle depth of only 20% C_N. This indicates that the capacity fade can decrease when the time the cell spends in the high SoC region shortens.

The capacity curves depicted in Figure 6c reveal that a higher capacity fade has occurred at the low operating temperature of 10°C. The step at 800 EFC in the capacity graph of the cell cycled with a low cycle depth at high SoC will be explained in a subsequent section. Particularly for higher cycle depths, the capacity fade

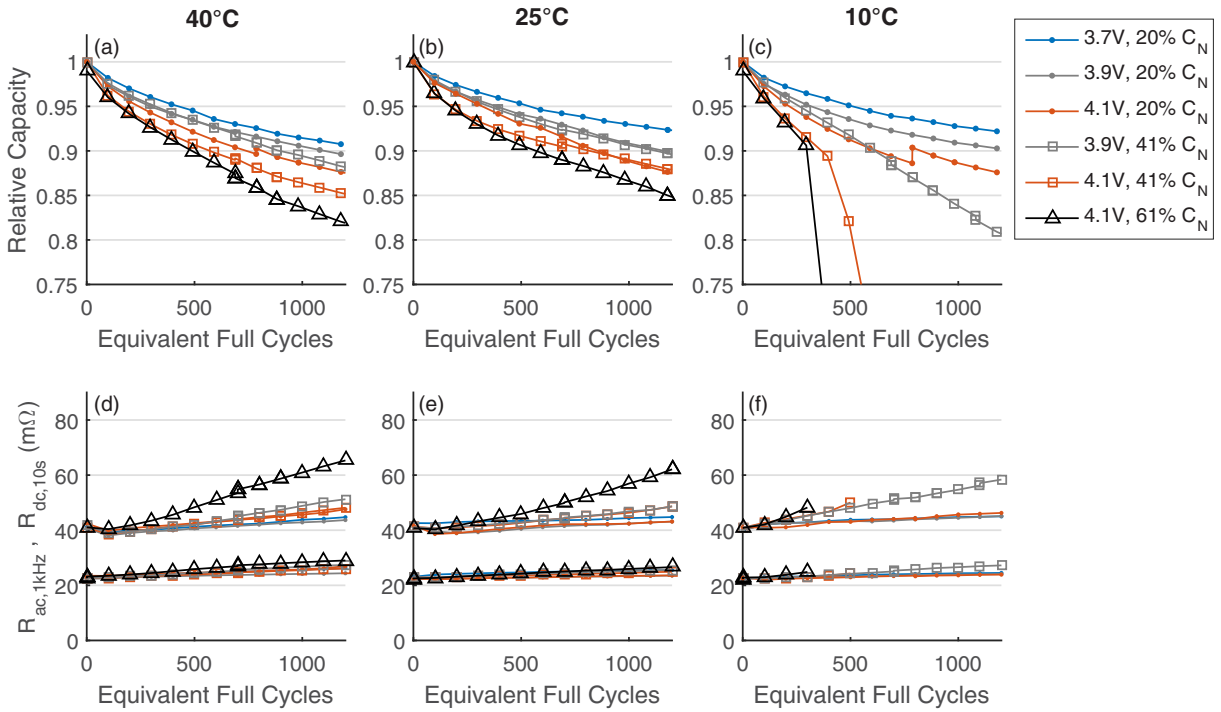


Figure 6. (a-c) Capacity fade and (d-f) resistance increase for up to 1200 EFC with unrestricted regenerative braking, different charging voltages, and depths of discharge of 20%, 41%, and 61% C_N. (For 61% C_N at 40°C and 10°C, the relative capacity curves were shifted 0.01 down, since the cells had already performed several cycles in another study so that their absolute capacity and the initial capacity fade were somewhat lower.)

increases markedly. There have even been premature failures. The cells which were charged to 4.1 V and discharged with a cycle depth of 61% or 41% C_N both failed in the checkup sequence when the cells were charged to 4.2 V. No voltage was available at the terminals any longer. This occurred after 400 EFC and 600 EFC, respectively. An opening of the cells and a continuity check at the positive pole of the cell confirmed a tripping of the current interrupt device (CID), which occurs in case of an overpressure inside the cell.[38] Overpressure inside the cell is typically a result of gaseous reaction products from electrolyte decomposition.[39] The pressure rise can be explained by intensified lithium plating and subsequent side reactions with the electrolyte. Waste products from these side reactions can also lead to local pore clogging, which further accelerates the formation of newly plated lithium and leads to a disproportionate capacity fade.[9,40]

Such a premature failure also occurred after 700 EFC at 10°C for the cell with a charging voltage of 4.1 V and a cycle depth of 25% C_N without regenerative braking. This demonstrates that it is not the short-time recharging from regenerative braking which causes the gassing side reactions. Instead, increasing cycle depths and high charging voltages accelerate the degradation at low temperature.

At all three temperatures examined, the capacity fade has increased considerably with higher cycle depths, in particular at high SoC. The best cycle life has always been obtained for cycling with low cycle depths in the low SoC region, i.e., below 50% SoC.

Resistance increase.—The resistance developments, depicted in Figures 6d–6f, show the changes in the internal cell resistances. The changes in $R_{ac,1kHz}$, representing changes in the electrolyte conductivity, increase with higher temperature and larger cycle depths. However, the absolute changes remain small compared to the changes in $R_{dc,10s}$, which also comprises the rising charge transfer resistances of the cathode. Increasing charge transfer resistances are typical for aged NCA cathodes.[41,42] With higher cycle depths, larger volume changes occur, which cause mechanical stress and lead to microcracks in the cathode active material.[42,43]

At 40°C, $R_{ac,1kHz}$ rises by 2–6 mΩ whereas $R_{dc,10s}$ exhibits increases of up to 25 mΩ for the cycle depth of 61% C_N. The initial reduction of $R_{dc,10s}$ due to decreasing anode resistances is also observed for the larger cycle depths. Comparable results were also observed for 25°C. The two cells with a cycle depth of 41% C_N exhibit similar resistance changes, although they have been cycled at different SoC regions. This confirms that the resistance increase depends mainly on the cycle depth and less on the actual SoC interval.

At 10°C, only the cells with a cycle depth of 20% C_N and the cell with a cycle depth of 41% C_N in combination with the medium charging voltage were able to perform 1200 EFC. For the latter, $R_{ac,1kHz}$ has increased by 5 mΩ. For the cells with the low cycle depth of 20% C_N, $R_{ac,1kHz}$ has increased by only 2 mΩ.

Long-term cycling at 25°C.—To analyze the long-term cycle aging of the lithium-ion cells, the test procedures for the cells cycled at 25°C with unrestricted regenerative braking were continued to 2000 EFC, which correspond to a driven distance of about 200,000 km. Figure 7 illustrates the capacity fade and the resistance increase for the different charging voltages and cycle depths.

Capacity fade.—The strong dependency of capacity fade on cycle depth is also observed in the long-term aging data depicted in Figure 7a. After 2000 EFC, the cells with a cycle depth of only 20% C_N exhibit a capacity fade of 10% at low SoC, 12% at medium SoC, and 15% at high SoC. The cells with a cycle depth of 41% C_N exhibit a capacity fade of 13.5% for the medium charging voltage and 15.5% for the high charging voltage. The largest capacity fade of more than 20% was observed for cycling with 61% C_N cycle depth.

After 1200 EFC, the cells cycled with the low cycle depth of 20% C_N at medium or high SoC feature a step in the capacity curve of 2–3 percentage points. This corresponds to an interruption of the aging study lasting more than 5 months. The long pause led to a considerable capacity recovery, which will be further analyzed in the subsequent result section.

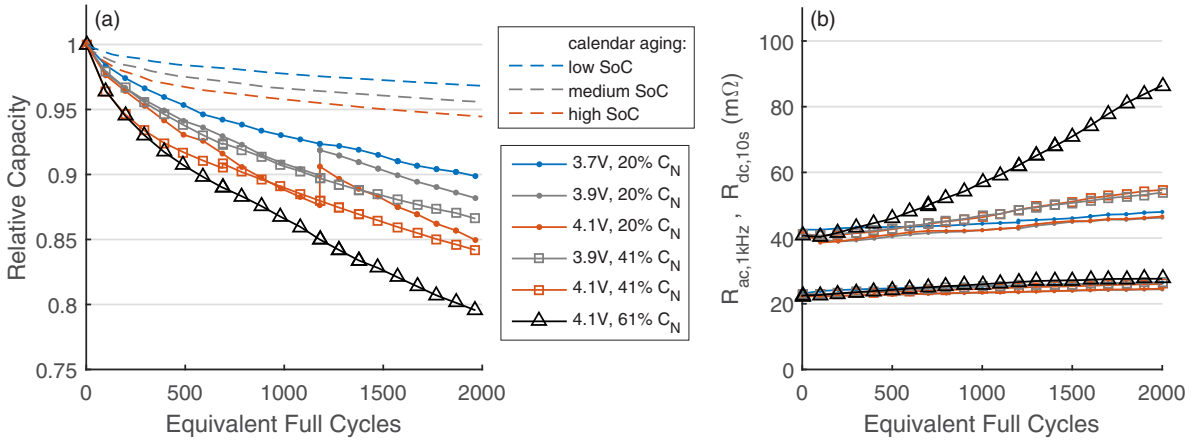


Figure 7. (a) Capacity fade and (b) resistance increase for long-term cycling at 25°C with unrestricted regenerative braking for different charging voltages and cycle depths. The dashed lines represent capacity fade owing to calendar aging at low, medium, and high SoC, with the same time scaling as in Figure 4.

The difference between the capacity fade of the cycled cells and the calendar aging curves reveals that the additional capacity fade owing to cycle aging for 200,000 km at 25°C can be estimated at 7–16%. The lowest cycle aging is obtained for cycling with low cycle depth of 20% C_N in the low SoC region below 50% SoC. With higher SoC and particularly with increasing cycle depth, the capacity fade aggravates. In this study, the highest cycle aging is observed for the largest cycle depth of 61% C_N.

This demonstrates that lithium-ion battery aging strongly correlates with cycle depth. Hence, a direct recharging after partial discharging – even after short driving distances – reduces the cycle depth and might be beneficial to prolong battery life, although the battery spends more time at a higher SoC. Thus, an appropriate compromise between cycle aging and calendar aging has to be found.

Resistance increase.—Also the resistance increase aggravates with higher cycle depth, which can be observed particularly in the $R_{dc,10s}$ values. For the cells with identical cycle depth, Figure 7b exhibits similar trends in the resistance increase. After 2000 EFC, the $R_{dc,10s}$ has increased by 46 mΩ for the cycle depth of 61% C_N. This means that the resistance has more than doubled. For the cycle depth of 41% C_N, the increase of $R_{dc,10s}$ is 14 mΩ and for the low cycle depth of 20% C_N, it is below 6 mΩ. The corresponding EIS measurements confirmed that the resistance increase was again dominated by increasing charge transfer resistances, which can be ascribed to the aged NCA cathode.

Regeneration effects.—In the previous section, capacity recovery has been observed after an interruption of the cycle aging study of several months. The capacity recovery occurred for the cells cycled with low cycle depth of 20% C_N at the high and medium SoC level. To further investigate the capacity recovery during interrupts, the capacity development is analyzed over time to identify the origin of the reversible capacity fade.

Capacity fade over time.—Figure 8 depicts the capacity development over time – instead of over EFC, as was the case in Figure 7a. The markers represent the checkups, the continuous lines represent cycling periods, and the dotted lines indicate pauses with no cycling.

After the first 140 days, in which 500 EFC with a cycle depth of 20–25% C_N were tested for various combinations of charging voltage, temperature, and magnitude of regenerative braking, the cells with the intermediate levels of regenerative braking were stopped. Their test channels were then used to examine further load scenarios, particularly those with larger cycle depths. After 380 days, corresponding to 1200 EFC for the cells cycled from the very beginning and to 700 EFC for the cells newly started after six months, the aging study was interrupted for more than five months. During this long interruption, the cells were stored at 10°C to minimize calendar aging. In all pauses, the cells were stored at 50% SoC, as this was the SoC of the cells at the end of each checkup.

When the study continued about 160 days later, an additional checkup was performed to determine the actual capacity before starting the new cycling sequences. These capacity measurements reveal substantial capacity recovery for the cells cycled with low cycle depth at high and medium SoC. This capacity recovery caused the steps in the capacity curves which have been presented earlier in Figure 7a. For cycling with low cycle depth at high SoC, Figure 6a and Figure 6c have also shown a capacity recovery at 40°C and 10°C. These two cells were also started later. As a consequence, the about 5 months of pause correspond to the regeneration at 800 EFC. All other cells depicted in Figure 6 were started at the very beginning and, thus, did not experience a longer pause within the 1200 EFC.

For the cells cycled with the low cycle depth of 20% C_N at low SoC, no capacity recovery has occurred during the test interruptions. There is also no capacity recovery for the large cycle depth of 61% C_N and for the cycle depth of 41% C_N in combination with the medium charging voltage of 3.9 V. This indicates that all cells that reach the low SoC domain during the cycling procedure do not exhibit capacity recovery. For all cells cycled within the high or medium SoC range only, which means between 90% and 50% SoC only (see Figure 2), a capacity recovery has been observed.

Origin of the reversible capacity fade.—To determine the origin of the capacity recovery, a recently introduced coulometry method,

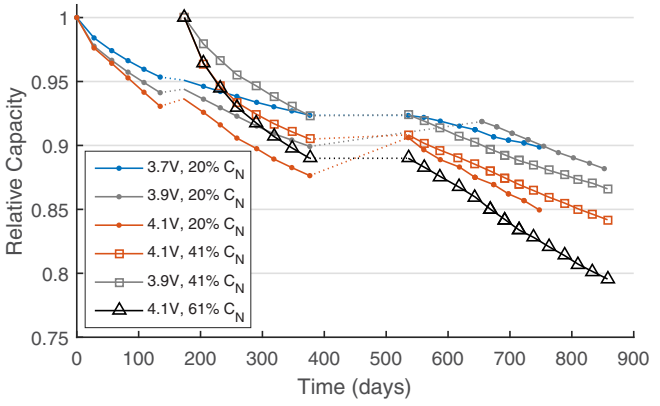


Figure 8. Capacity development over time. The markers represent checkups, continuous lines represent cycling periods at 25°C, and dotted lines indicate storage periods. This demonstrates the capacity recovery of cells cycled in higher SoC regions only.

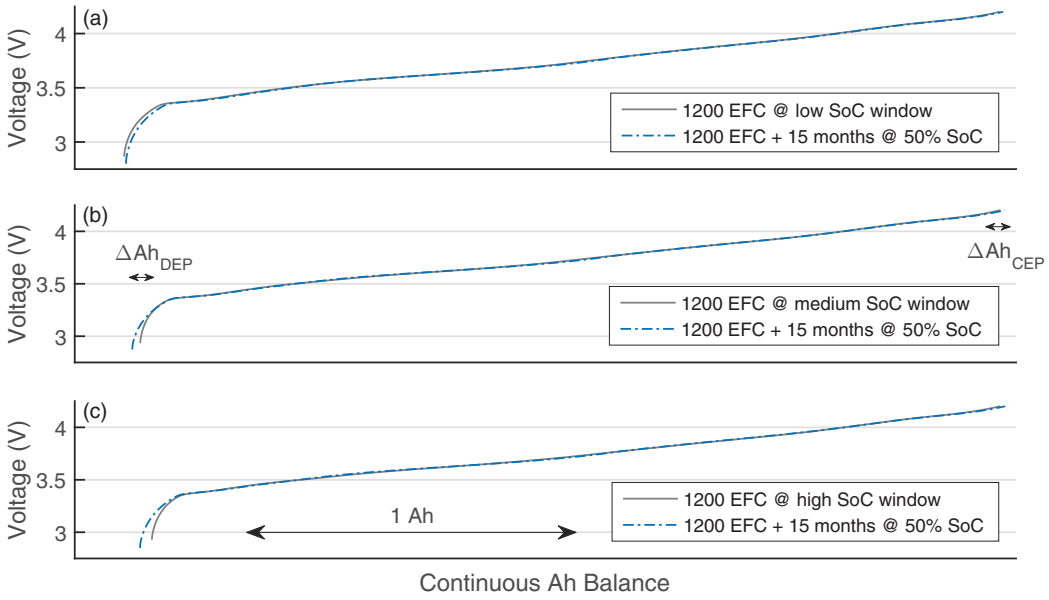


Figure 9. Capacity recovery after 15 months of storage at 50% SoC examined by coulomb tracking for the cells cycled 1200 EFC without regenerative braking in the (a) low, (b) medium, and (c) high SoC window with a depth of discharge of 25% C_N. The x-axes are scaled identically for all three cells.

termed coulomb tracking,[26] is employed. The main idea behind coulomb tracking is that all tests performed with a battery cell throughout a selected time period are brought in a consecutive order and a continuous ampere-hour balance over all tests is computed. Based on this continuous ampere-hour balance, charging end point (CEP) slippage and discharging end point (DEP) slippage is evaluated for an identical charging or discharging sequence which is included in several measurement datasets. This enables the identification of anodic and cathodic side reactions which modify the operating window of a lithium-ion cell and, thus, alter the amount of cyclable lithium.

For the coulometry evaluations, the cells cycled with no regenerative braking at 25°C were examined. The cycling procedure of these cells was discontinued after 1200 EFC. Before and after 15 months of storage at 50% SoC, an identical charge-discharge sequence was performed with these cells at 25°C, in which the cells were CCCV charged to 4.2 V with 700 mA, discharged and recharged with 100 mA to 2.5 V and 4.2 V, respectively, and finally discharged to 50% SoC. Based on these two measurements, coulomb tracking was performed and the end point slippages can be analyzed. Figure 9 shows the low-current charging sequences of the two measurements versus the continuous ampere-hour balance. Figure 9b and Figure 9c reveal a considerable DEP slippage for those cells that were cycled in the medium and high SoC cycling window before the long storage period. Table II lists the explicit slippages of both endpoints for all three cells. This exhibits that there are comparatively small differences among the CEP slippage values but marked differences among the DEP slippage values. Table II also demonstrates that the capacity recovery of up to 1.8% C_N mainly originates from the DEP slippage, which represents a change in the amount of cyclable lithium owing to the anode.[26]

A mechanism that explains the observed recovery of cyclable lithium was described by Gyenes et al.,[44] Lewerenz et al.,[45] and Wilhelm et al.:[46] Overhang areas of the anode, which face no cathode counterpart, can become a location of inaccessible lithium. Due to potential gradients between the anode active areas, which face a cathode counterpart, and the overhang areas, lithium can migrate into overhang areas and become inaccessible for the regular charge-discharge cycling. When a cell is operated at high SoC most of the time, the potential of the anode active area mostly lies within the low voltage plateau of LiC_6/LiC_{12}.[47,48] This leads to a larger amount of lithium moving into the overhang areas and a lower potential in these areas. As a consequence, the amount of cyclable lithium decreases. When the cells are then stored for several months at 50% SoC, the potential of the anode active area lies within the medium graphite voltage plateau. As a consequence, the potential gradients reverse and lithium from the overhang areas moves back into the active areas, where it becomes accessible again. Thus, the amount of cyclable lithium increases and the cells provide a higher capacity again.

This demonstrates that the battery SoC during longer rest periods of an EV can influence the available capacity notably. In general, EVs should not be operated at high SoC only, but be kept at a medium or low SoC from time to time to recover inaccessible lithium from the anode overhang areas.

Comparison of dynamic discharging and constant-current discharging.—The US06 driving load profiles used in this study changed the load current every second. To determine the impact of these frequent load changes on battery aging, additional cells were tested with a CC discharging procedure, which featured the same cycle depth and the same mean discharging current as the load profile with unrestricted regenerative braking. The dynamic driving load profile contained load currents between –5.5 A and +4.5 A with a mean discharging current of –1.714 A. Four repetitions of the driving

Table II. Slippage of the discharging end point (ΔAh_{DEP}) and the charging end point (ΔAh_{CEP}) after the pause of 15 months. Positive values represent a right-shift whereas negative values represent a left-shift of the end point. Both end point slippages contribute to the change in the available capacity (ΔAh).

Load Condition before Pause	ΔAh_{DEP}	ΔAh_{CEP}	$\Delta Ah = \Delta Ah_{CEP} - \Delta Ah_{DEP}$
Cycling at low SoC window	+5.4 mAh	+9.6 mAh	+4.2 mAh = +0.15% C_N
Cycling at medium SoC window	–24.2 mAh	+15.0 mAh	+39.2 mAh = +1.4% C_N
Cycling at high SoC window	–35.8 mAh	+15.8 mAh	+51.7 mAh = +1.8% C_N

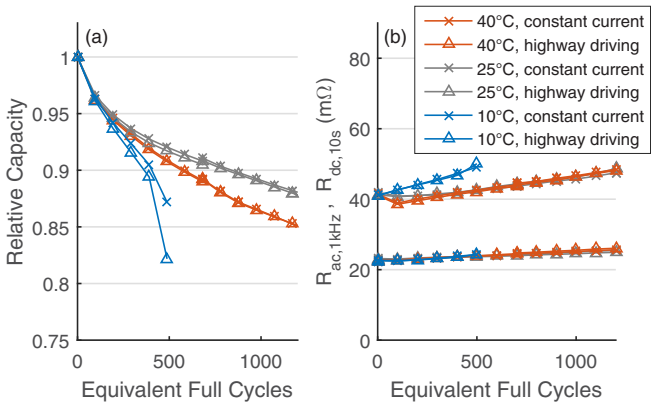


Figure 10. Comparison of battery degradation for the dynamic driving load profile and constant-current discharging with the same cycle depth of 41% C_N and the same mean discharge current of –1.7 A. (a) Capacity fade and (b) resistance increase.

load profile led to a cycle depth of 41% C_N. For both load cases, the charging procedure was CC charging with 700 mA to 4.1 V. Figure 10 compares the capacity fade and the resistance increase of the two load scenarios at different operating temperatures.

Capacity fade.—Figure 10a shows that the capacity fade of the two cells cycled at 40°C is similar. At 25°C, the capacity fade is somewhat lower for discharging with the constant current. However, the differences are small and the additional capacity fade due to the dynamic driving load profile with its multiple peak pulse loads lies below 0.4 percentage points. After 700 EFC, where the aging study has been interrupted for more than 5 months, a similar capacity recovery has been observed for both load conditions. As described above, the differences in capacity fade between 25°C and 40°C can be ascribed largely to the increasing calendar aging with higher temperature.

For cycling at 10°C, Figure 10a exhibits a substantially faster degradation than for 25°C and 40°C. Moreover, there is a considerable difference in capacity fade between both load scenarios. The dynamic driving load profile has led to a faster capacity fade. At this low temperature, both cells exhibit an accelerated degradation after 400 EFC. As premature failures had been expected after the rapid degradation throughout the last 100 EFC, the cycling of these cells was stopped after 500 EFC.

Resistance increase.—Figure 10b illustrates the resistance increase for the different combinations of load scenario and temperature. For all three operating temperatures, the resistance increase is similar for both load scenarios. From these findings, it is deduced that lithium-ion battery aging exhibits a certain low-pass behavior, which means that the resistance increase is driven by the mean load current and the cycle depth, and not by short peak loads. Such a low-pass behavior was also observed in our preceding aging study on charging protocols for lithium-ion batteries.[6]

Conclusions

The cycle aging study presented in this paper has provided valuable insights into battery degradation under EV driving operation.

Cycle aging decreases with temperature.—Comparing different operating temperatures has revealed that the capacity fade from cycling that occurs in addition to calendar aging decreases considerably with temperature. This is opposed to calendar aging which increases with higher temperature. At low temperature, lithium plating becomes the driving force for capacity fade, as it entails a loss of cyclable lithium. When cycling at 10°C, accelerated degradation has been observed, which even led to a tripping of the CID in several cases. This has been caused by a rise of internal cell pressure, which can be related to gaseous reaction products from side reactions between plated lithium and the electrolyte. Overall, the study has demonstrated that a warm battery is beneficial during driving operation. Low-temperature operation is still a critical issue for lithium-ion batteries.

Regenerative braking improves cycle life.—The comparison on different levels of regenerative braking has demonstrated that for a typical driving load profile, the short recharging periods during braking do not increase battery degradation – even at low battery temperatures of 10°C. By contrast, reduced degradation was observed with higher levels of regenerative braking, in particular at high SoC and at low temperature, which are operating conditions that typically intensify lithium plating. The reduced degradation has been ascribed to the reduced depth of discharge when the battery is partially recharged during braking periods, which leads to shorter durations of the subsequent recharging periods. It has been shown that the capacity fade depends more on the amount of charge recharged at the charging station than on the overall charge throughput. Thus, a high level of regenerative braking is beneficial for an EV. However, further studies employing even more aggressive braking actions with higher recharging currents should be performed.

Cycle depth as dominant factor for cycle life.—Cycle depth has been identified as a major influencing factor for battery degradation. With higher cycle depths, capacity fade and resistance increase are aggravated. In particular the resistance increase has revealed a strong dependency on cycle depth and only a small dependency on SoC. Moreover, the resistance increase has mainly been caused by rising charge transfer resistances of the NCA cathode. The increasing resistances lead to higher losses, which have to be considered when designing the cooling systems for EV batteries.

Recovery effects due to stored lithium in the anode overhang areas.—Capacity recovery effects without changes in the internal resistances were observed during longer pauses of several months for cells cycled at high SoC only. Coulomb tracking analysis has revealed a left-shift of the discharging end point after the pauses, which means that more lithium could be discharged from the graphite anode again. Such a reversible capacity fade occurs due to lithium moving into the overhang areas of the anodes, which face no cathode counterpart. When cells are kept a high SoC, corresponding to the plateau of lowest anode potential, for a longer time, a considerable amount of lithium migrates into the side areas and becomes inaccessible during the cycling sequences. To avoid an accumulation of inaccessible lithium, prolonged times at high SoC should be avoided and longer times at medium or low SoC help to recover parts of the inaccessible lithium by reversing the potential gradients between active areas and overhang areas of the anode active material.

Degradation under dynamic loads mostly similar to constant loads.—The comparison between the dynamic driving load profile and a constant load profile with identical mean current and cycle depth has revealed that degradation is similar in most cases. Only for the low temperature of 10°C, the driving load profile has led to faster degradation than the continuous load. This result, together with the results from different magnitudes of regenerative braking, has demonstrated that lithium-ion batteries exhibit a certain low-pass behavior, which means that the degradation is determined largely by the average load and not by short peak load events. Only for the low temperature of 10°C, fluctuating discharging currents have appeared to be more detrimental than a constant discharging procedure. Thus, for warm and moderate battery temperatures, conventional CC cycling appears appropriate for cycle life testing of EV batteries. For cold temperatures, however, the explicit dynamics of the load profile should be considered.

Operating conditions for long cycle life.—For cycle life estimations for an EV, cells have been cycled up to 2000 EFC which is far

beyond the USABC development goals for future EV batteries of 1000 cycles.[49] The 2000 EFC could be achieved at 25°C with less than 20% capacity fade for cycle depths up to approx. 60%. Moreover, it has to be considered that for softer load patterns than the constant US06 highway driving, battery aging is expected to be lower. Overall, the best cycle life has been obtained for a low cycle depth in combination with a low average SoC. In general, avoiding the SoC regimes of lowest anode potential reduces calendar aging and also the susceptibility to lithium plating.

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